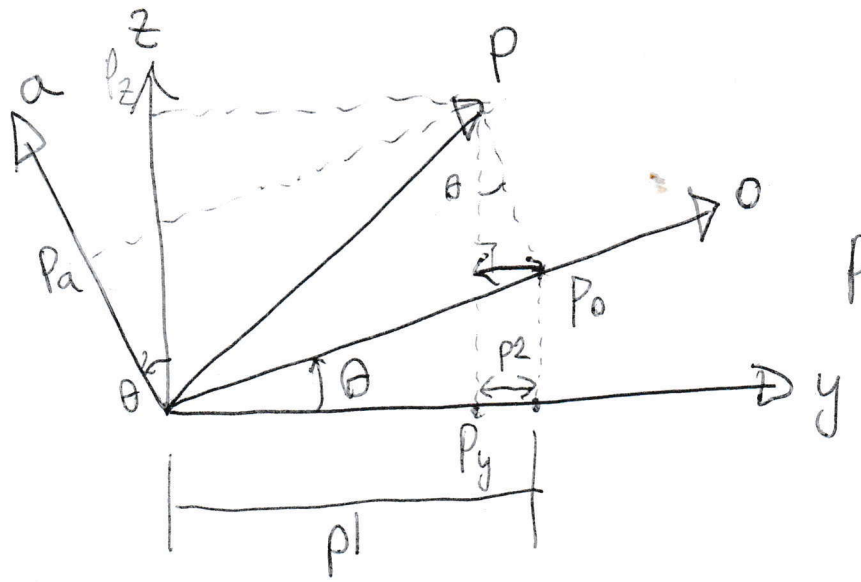
$$\bar{P} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 6 \\ 2 \\ 0 \end{bmatrix} = ? \qquad \begin{bmatrix} 3/7 \\ 6/7 \\ 2/7 \\ 0 \end{bmatrix} = \text{UNIT VECTOR.}$$

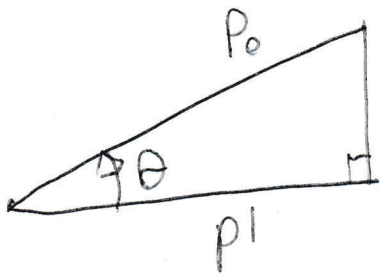
ROTATION ABOUT X-AXIS

$P_x = P_n$

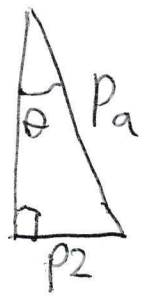
P_y



$P_y = p1 - p2$



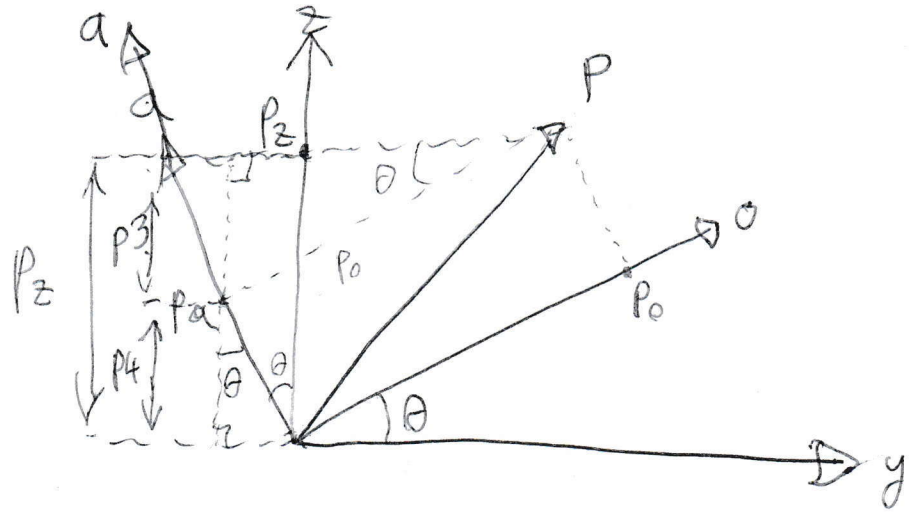
$\therefore p1 = P_o \cos\theta$



$\therefore p2 = P_a \sin\theta.$

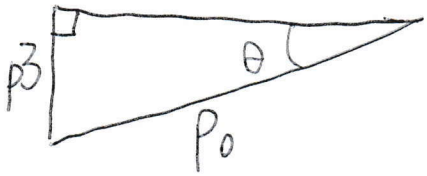
$$P_y = P_o \cos\theta - P_a \sin\theta.$$

P_z



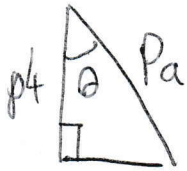
$$P_z = P_3 + P_4$$

$p3$:



$$\therefore \quad p3 = P_o \sin\theta$$

$p4$:



$$\therefore \quad p4 = P_a \cos\theta.$$

$$P_z = P_3 + P_4$$

$$= \underline{\underline{P_o \sin\theta + P_a \cos\theta}}$$

$$\left.\begin{aligned} P_x &= P_n. \\ P_y &= P_o \cos\theta - P_a \sin\theta \\ P_z &= P_o \sin\theta + P_a \cos\theta \end{aligned}\right\} \quad \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} P_n \\ P_o \\ P_a \end{bmatrix}$$

$$\underline{P}_{xyz} = \underline{T}\ \underline{P}_{noa}.$$